

does the Internet decipher which computer you want to contact to receive data from or send data to? We'll explain all...

Step 1: When a specific URL needs to be contacted, the network needs to match the URL to the actual IP address (the alphabetical gobbledygook to the safe, meaningful language of numbers). Two things could happen now:

- Your Web browser first goes to the local name server (a DNS server) maintained by your ISP or company, and checks if the IP address is on the same network as you. If it finds it right there, it resolves the URL with the corresponding IP address, and sends the true IP address to your computer.
- Or, the information you have requested is not on your local network. The local name server will try to get the information from some other DNS server on the Internet. The local name server contacts the root domain server, which supplies the local name server with details about which Primary Name Server and Secondary Name Server actually hosts the URL you requested. The local name server (the DNS server of your ISP) now contacts the primary name server. If it still can't find the URL there, it contacts the secondary name server. It now sends the information about the true IP address of the URL back to your browser.

Step 2: Either way, your Web browser now has the IP address of the URL you want to access and can use it to contact the proper site.

Step 3: The request is broken up into IP packets by TCP.

Step 4: These packets go to an internal router, a router that may not be on the Internet at all but in your own network (LAN or WAN)—which checks out the address.

Step 5: The internal router now figures out whether the request is targeted to another computer on the same network or one outside the network. If it finds the destination on the same network, it forwards your request to it. Or else it directs your request to an external router on the Internet.

Step 6: The router usually uses “dynamic routing” and allows the packets of data to have multiple routes to reach the destination (the host computer). This enables the router to change the way in which the information can be routed if there are any glitches in the system preventing smooth passage of data, such as traffic or hardware breakdowns.